

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Pujitha

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.
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RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose,	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article

		concepts, and arguments			
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

TASK 1 :

(a) Full Citation

Authors: Leo Breiman

Title: *Random Forests*

Journal: Machine Learning

Year: 2001

DOI: 10.1023/A:1010933404324

Domain/Application Area:

Machine Learning, Data Mining, Classification and Regression.

Specific ML Problem Addressed:

The article addresses the problem of developing an accurate and robust machine-learning method for classification and regression, while reducing overfitting and improving prediction performance when working with large and complex datasets.

(b) Key Objective and Research Gap

The main objective of the article is to introduce Random Forests, an ensemble learning method that combines multiple decision trees to obtain more accurate and reliable predictions. Before this work, individual decision trees were widely used because they were easy to understand and capable of handling different types of data. However, single decision trees could be sensitive to changes in training data and could suffer from overfitting. The research gap was the need for a learning method that could maintain the flexibility of decision trees while providing better accuracy, stability, and resistance to overfitting. To address this gap, the author proposed Random Forests, which construct many decision trees using randomly selected training samples and random subsets of features. The predictions from these trees are then combined to produce the final result. The study demonstrated that this approach can achieve high classification accuracy and is effective for handling high-dimensional datasets. Thus, the article filled the research gap by providing a powerful ensemble technique that improves the reliability and generalization ability of decision-tree-based learning.

TASK 2 :

(a) ML Technique and Dataset

ML Technique Used

The main machine-learning technique used in the article is Random Forest, an ensemble method based on multiple Classification and Regression Trees (CART). Instead of depending on a single decision tree, Random Forest constructs a large number of trees and combines their predictions.

Two important ideas are used:

1. **Bagging:** Different bootstrap samples are created from the training data, and a separate decision tree is trained on each sample.
2. **Random Feature Selection:** At each node of a tree, only a randomly selected subset of features is considered for splitting. This reduces the similarity between trees and improves the overall performance of the forest.

For classification, the final prediction is generally obtained through majority voting among the trees. For regression, the predictions of the individual trees are averaged.

Dataset Used

The article does not rely on only one dataset. Breiman evaluates Random Forests using multiple real-world and simulated datasets to demonstrate its performance for classification and regression.

The experiments include datasets from areas such as image recognition, medicine, biology, and other classification applications. The datasets contain different numbers of observations and input variables, allowing the method to be tested under different conditions.

The article also uses datasets such as waveform data and other benchmark datasets to compare Random Forests with other machine-learning techniques.

The data were prepared according to the requirements of each experiment. Bootstrap samples were generated from the available observations, and random subsets of variables were selected during tree construction. An important feature of Random Forest is that an independent test set is not always required: the out-of-bag (OOB) observations can be used to estimate prediction error.

(b) Mapping to CO 4 Topics

CO 4 Topic	Connection with the Article
Inductive Learning	The model learns general patterns from training examples and uses them to predict unseen data.
Decision Trees	Random Forest is directly built using multiple decision trees.
Statistical Learning	The method uses bootstrap sampling, randomization, aggregation, and statistical estimation to improve prediction.
Reinforcement Learning (RL)	Not directly applicable. The article does not use states, actions, rewards, or Q-learning.

Methodological Strength

A major strength is robustness and generalization. By combining many randomized decision trees, Random Forest reduces the risk of overfitting compared with a

single decision tree. The use of out-of-bag observations also provides an efficient way to estimate prediction error without requiring a separate validation set.

Methodological Limitation

One limitation is that Random Forests can become computationally expensive and difficult to interpret when a very large number of trees are constructed. Unlike a single decision tree, it is difficult to clearly explain the complete decision-making process of hundreds of trees. Therefore, although the method provides strong predictive performance, its interpretability is lower than that of a simple decision tree.

TASK 3 :

(a) Key Results

Breiman evaluated Random Forests on several classification and regression datasets and compared their performance with other machine-learning methods. The main finding was that Random Forests generally achieved low generalization error and were competitive with or better than many existing classifiers.

The article particularly emphasizes the stability of Random Forests and their ability to maintain good accuracy even when the number of trees is large. The out-of-bag (OOB) error was used as an important estimate of generalization performance.

Method / Model	Main Result Reported	Evaluation
Single Decision Tree	Higher variability and greater susceptibility to overfitting	Relatively less stable
Bagging	Reduced variance compared with a single tree	Good performance
Random Forest	Generally lower error and strong classification performance	Best/among the best in many experiments

Method / Model	Main Result Reported	Evaluation
Randomized Trees	Competitive performance	Useful benchmark

The article does not report modern metrics such as F1-score, AUC-ROC, or reinforcement-learning reward convergence, because the study predates their widespread use in this context. Its principal evaluation measure is generalization error/error rate, along with comparisons against other classifiers.

(b) Critical Evaluation

The reported results are generally realistic because the experiments were conducted on multiple datasets rather than relying on only one dataset. The use of OOB error is also a methodological strength because it provides an internal estimate of prediction performance.

However, some limitations should be considered. The datasets used in the study may not represent all modern real-world datasets, especially highly imbalanced, streaming, or extremely high-dimensional datasets. Performance can also depend strongly on the selected datasets and experimental settings. Therefore, strong results on benchmark datasets do not automatically guarantee the same performance in every practical application.

Potential biases may arise from dataset selection, class distribution, feature quality, and differences between benchmark data and real-world data. In addition, the original evaluation does not provide some metrics commonly expected today, such as precision, recall, F1-score, confidence intervals, and AUC-ROC.

Additional Experiments That Would Strengthen the Findings

The study could be strengthened by:

1. Testing Random Forest on larger and more diverse modern datasets.
2. Reporting accuracy, precision, recall, F1-score, and AUC-ROC for classification tasks.
3. Performing repeated cross-validation and reporting confidence intervals.
4. Testing performance on highly imbalanced datasets.

5. Comparing Random Forest with modern ensemble methods such as Gradient Boosting, XGBoost, and LightGBM.
6. Measuring training time, prediction time, memory usage, and scalability.
7. Conducting feature-importance and interpretability experiments.

(c) Real-World Applicability

Random Forests can be deployed in many real-world applications because they can handle both numerical and categorical features and are relatively robust to noisy data.

Possible Applications

- **Healthcare:** disease-risk prediction and patient classification.
- **Banking:** credit-risk assessment and fraud detection.
- **Cybersecurity:** intrusion and malware detection.
- **Manufacturing:** fault and predictive-maintenance classification.
- **Agriculture:** crop-disease and crop-condition prediction.
- **Marketing:** customer segmentation and churn prediction.
- **Environmental monitoring:** classification of environmental conditions.
- **Image and biological data analysis:** pattern and feature classification.

Challenges in Industry

1. Data availability:

Large amounts of high-quality, representative, and correctly labelled data may not always be available.

2. Computational requirements:

A forest containing many trees can require considerable memory and computation, particularly for very large datasets.

3. Data drift:

Real-world data can change over time. A model trained on historical data may lose accuracy when customer behaviour, environmental conditions, or business processes change.

4. Interpretability:

Although feature importance can be calculated, understanding the exact decision process of hundreds of trees is more difficult than interpreting one decision tree.

5. Ethical and privacy concerns:

Applications involving healthcare, finance, or personal information must address privacy, fairness, security, and potential algorithmic bias.

Overall Evaluation

Overall, the article provides strong evidence that Random Forests are accurate, robust, and effective ensemble-learning models. Its major contribution is demonstrating that combining randomized decision trees can significantly improve predictive performance and generalization. However, modern evaluation using diverse datasets, additional performance metrics, computational benchmarks, fairness analysis, and comparisons with current state-of-the-art models would provide stronger evidence for present-day industrial deployment.

TASK 4 :

(a) Three Key Insights and Connection to CO 4

Insight 1: Combining Multiple Decision Trees Improves Prediction

I learned that using a group of decision trees can provide better and more stable predictions than relying on a single tree. Random Forest combines the outputs of many trees through ensemble learning.

CO 4 Connection: This directly relates to Decision Trees and Inductive Learning. A decision tree learns rules from training examples, while Random Forest improves this learning process by combining multiple learned models.

Insight 2: Randomization Helps Reduce Overfitting

A major insight from the article is that Random Forest introduces randomness through bootstrap sampling and random feature selection. Because the individual trees are not identical, their errors can partially cancel when their predictions are combined.

CO 4 Connection: This relates to Statistical Learning, particularly sampling, variance reduction, and generalization. It helped me understand how statistical techniques can improve the performance of machine-learning models.

Insight 3: Out-of-Bag Error Can Evaluate the Model

I learned that Random Forest can use observations that were not selected in a particular bootstrap sample as out-of-bag (OOB) data to estimate prediction error. This provides a useful internal evaluation mechanism.

CO 4 Connection: This is related to Inductive Learning and Statistical Learning, because the model learns from training examples and evaluates how well the learned pattern generalizes to unseen observations.

Overall, the article helped me understand that ensemble learning, statistical randomization, and proper evaluation are important for building reliable machine-learning systems. It also clarified that Random Forest belongs mainly to Decision Tree and Statistical Learning topics rather than Reinforcement Learning.

(b) Proposed Extension

If I were to extend this research, I would conduct an experiment using Random Forest for real-time healthcare disease-risk prediction with a modern patient dataset.

The experiment would compare the standard Random Forest with an optimized Random Forest using feature selection and hyperparameter tuning. The model could be evaluated using accuracy, precision, recall, F1-score, AUC-ROC, training time, and inference time.

This experiment is feasible because publicly available healthcare datasets can be used without requiring the collection of new patient data. Feature-selection methods could remove irrelevant attributes, while hyperparameter optimization could identify a suitable number of trees, tree depth, and feature-selection strategy.

The expected impact would be improved predictive performance and reduced computational cost. More importantly, the experiment could demonstrate whether the Random Forest approach proposed in the original article remains effective for a modern, high-dimensional, real-world application. Adding explainability methods

such as feature importance or SHAP would also help healthcare professionals understand the factors influencing predictions.